

MEDICAL DEVICE MARKETING

UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

1. Project Definition

Introduction

A **project** is a planned set of activities carried out to achieve a specific objective within a defined **scope, time, cost, and quality**.

In medical device development, a project may involve designing, developing, testing, validating, obtaining regulatory clearance/approval, and launching a new medical device.

Example

Development of an **IoT-based patient monitoring system** can be considered a project with:

- Objective – Develop a real-time patient monitoring system
- Duration – 12 months
- Budget – ₹20 lakh
- Team – Biomedical engineers, software engineers, clinicians and marketing personnel
- Deliverable – Tested and validated prototype/product

Major elements of project definition

1. **Project title**
2. **Problem statement**
3. **Objectives**
4. **Scope**
5. **Deliverables**
6. **Target customers**
7. **Timeline**
8. **Budget**
9. **Resources**
10. **Quality requirements**
11. **Regulatory requirements**
12. **Risk management**

Importance

A clear project definition helps the team understand **what has to be developed, why it is required, who will use it, how much it will cost, and when it should be completed**.

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- Meet applicable regulatory requirements
- Are appropriately manufactured
- Have appropriate labeling
- Continue to be monitored after marketing

Step 1 – Identify the Intended Use

The first step is to clearly define:

- What the device does
- Who will use it
- Which patients will use it
- What disease/condition it addresses
- Whether it diagnoses, treats, prevents, monitors, or supports a condition

The intended use and indications for use are important because they influence the device's classification and regulatory pathway.

Step 2 – Determine Device Classification

FDA generally classifies medical devices into:

Class I – Low risk

Generally subject to **General Controls**.

Examples may include certain low-risk devices.

Class II – Moderate risk

Subject to **General Controls + Special Controls**.

Class III – High risk

Subject to **General Controls + Premarket Approval (PMA)** in applicable cases.

FDA states that classification is risk-based, with Class I generally representing the lowest risk and Class III the highest.

Class	Risk	Main Controls
Class I	Low	General Controls
Class II	Moderate	General + Special Controls
Class III	High	General Controls + PMA in applicable cases

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Step 3 – Identify the Appropriate Regulatory Pathway

The major pathways students should know are:

A. 510(k) – Premarket Notification

The manufacturer generally demonstrates that the new device is **substantially equivalent** to a legally marketed predicate device.

The comparison generally considers:

- Intended use
- Technological characteristics
- Performance
- Safety

FDA explains that a 510(k) is a pathway based on substantial equivalence to an appropriate legally marketed device.

Important: The correct term is generally "**FDA clearance**", not "FDA approval," for a 510(k).

B. De Novo Classification

The **De Novo pathway** can be used for certain novel devices for which there is no suitable predicate but whose risks can be appropriately controlled.

A successful De Novo request establishes a new classification for that device type. FDA identifies De Novo as a pathway for certain novel device types that have not previously been marketed in the United States.

C. Premarket Approval – PMA

PMA is the most stringent major premarket pathway.

It is generally associated with **Class III devices** requiring PMA.

The manufacturer must provide scientific evidence supporting the safety and effectiveness of the device. FDA describes PMA as the scientific and regulatory review used to evaluate the safety and effectiveness of applicable Class III devices.

Examples of devices that may require PMA include certain implantable and life-supporting devices.

D. HDE

Humanitarian Device Exemption (HDE) is associated with certain devices intended to benefit patients with rare diseases or conditions meeting the applicable statutory criteria.

Step 4 – Preclinical/Nonclinical Testing

Before clinical use or marketing, applicable devices may require testing such as:

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- Corrective and preventive actions
- Complaint handling
- Risk management
- Records

Step 7 – Prepare Regulatory Submission

The manufacturer prepares the appropriate regulatory submission.

Depending on the pathway, this may include:

- Device description
- Intended use
- Indications for use
- Design information
- Risk information
- Performance data
- Test results
- Clinical data where required
- Manufacturing information
- Quality-system information
- Labeling

Step 8 – Labeling

Medical device labeling is an important regulatory requirement.

FDA's general device labeling requirements are contained in **21 CFR Part 801**.

Labeling may include:

- Device name
- Manufacturer information
- Intended use
- Indications
- Instructions for use
- Warnings
- Precautions
- Contraindications

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- Installation instructions
- Maintenance information

Labeling must not provide misleading information or unsupported claims.

Step 9 – FDA Review

FDA reviews the submitted information according to the applicable pathway.

The review may involve:

- Technical evaluation
- Clinical evaluation
- Statistical evaluation
- Manufacturing/quality considerations
- Labeling review
- Risk assessment

FDA may request additional information where necessary.

Step 10 – Regulatory Decision

Depending on the pathway, the outcome may be:

- 510(k) clearance
- De Novo classification
- PMA approval
- HDE approval
- Exemption where applicable

Step 11 – Post-Market Requirements

FDA oversight does not end when a device reaches the market.

Post-market activities may include:

- Complaint handling
- Medical device reporting
- Corrective actions
- Recalls where necessary
- Post-market surveillance
- Manufacturing quality monitoring